University of London
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Ciao, I'm Fabio, I Am Very Interested in what Makes Romance Work

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Abstract

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Chern Number and Topological Insulators

Paragraph introducing the topics covered in this section. Points to hit:

- Periodic translational hamiltonians can be classified according to the chern number of their bands, where nonzer chern number implies the existence of gapless edge modes in open boundaries.
- This classification has been immensely successful for understanding a lot of materials eg: QHE, Spin Hall effect etc
- The classification relies on a notion of momentum space - in order to construct a manifold of states on which to define parallel transport.
- Thus it doesn't work for systems without k space - eg. open boundaries/disorder
- Local markers are an attempt at constructing a real space version of this
- We will explain all the prerequisites for constructing a local marker, as well as discussing their limitations.

Throughout this chapter and the next we will focus on non-interacting electrons on a lattice, thus let us begin by defining our Hilbert space in a first-quantised framework. We will work with the most general definition of a non-interacting tight-binding Hamiltonian in two dimensions. The Hilbert space $\mathcal{H}$ contains a set of N sites, where site j is at a position $\mathbf{r}_j$, and has η internal degrees of freedom. For example, our sites could correspond to atoms, and the states at each site could represent individual orbitals around an atom. We will not assume that our sites are arranged in a regular lattice – we will also want to model amorphous structures – thus, a state in this system can be written as

$$|\psi\rangle = \sum_j c_{\mathbf{r}_j,\alpha} |\mathbf{r}_j\rangle \otimes |\alpha\rangle \,, \tag{3.0.1}$$

where $c_{\mathbf{r}_j,\alpha}$ are a set of complex coefficients and $\alpha \in \{1...\eta\}$ labels the internal degree of freedom. A general Hamiltonian in this context takes the form

$$H = \sum_{j,k,\alpha,\beta} H_{j,k,\alpha,\beta} |\mathbf{r}_j\rangle \langle \mathbf{r}_k| \otimes |\alpha\rangle \langle\beta| \,. \tag{3.0.2}$$

If the system has translational symmetry $\{\mathbf{r}_j\}$ will define a regular lattice with periodic boundaries and the Hamiltonian is invariant under translations – we can of course diagonalise the Hamiltonian in a basis of k-states,

$$|\mathbf{k}\rangle = \frac{1}{\sqrt{N}} \sum_{\mathbf{r}_j} e^{-i\mathbf{k}\mathbf{r}_j} |\mathbf{r}_j\rangle . \tag{3.0.3}$$

For a finite system the k-states are quantised to $k_a = 2\pi n/L_a$, where L_a is the length of the system in direction $a \in \{x, y\}$, whereas in the case of an infinite system k can take any value in the interval $[0, 2\pi]$.

3.1 Abelian Berry Phase and the Chern Number

In this section we introduce the mathematical construction of Berry phase, Berry flux and Chern number. We will present these ideas in two different contexts. In §3.1.1 we introduce the simplest possible story that captures the essentials of Berry phase and Chern number: that of a discretised lattice of quantum states. Here wil will miss many of the details that make such concepts relevant to physical systems, however the picture is sufficiently simple as to provide a practical intuition for how to calculate these quantities. In §3.1.2, we will do things *properly*, by considering the effect on the band structure of some Hamiltonian that is changed adiabatically over some smooth parameter space. We will derive the exact form of the Chern number in this context and relate it to the TKNN invariant [1]. Finally, in §3.1.3 we give a brief example of such a calculation, demonstrating how these concepts apply to a two-state quantum system.

3.1.1 The Simplest Story – A Discrete Lattice of States

In quantum mechanics a state is defined as an equivalence class of vectors, where two states that differ only by a complex phase are considered physically equivalent,

$$|\psi\rangle \sim e^{-i\alpha} |\psi\rangle . \tag{3.1.1}$$

Thus, we seem to have a degeneracy built into our description of physics. This means that one can define a gauge transformation – multiplying a state with a complex phase – that is guaranteed not to affect the system in any 'real' measurable way. Any derived quantity must be invariant under gauge transformations in order to be considered physical. Thus, one would expect any quantity that depends on the state's phase to be unphysical.

Similarly, considering some arbitrary set of states in the same Hilbert space, $\{|\psi_j\rangle\}$, we may define a local gauge transformation where we apply a different phase to each individual state,

$$|\psi_j\rangle \rightarrow e^{-i\alpha_j} |\psi_j\rangle . \tag{3.1.2}$$

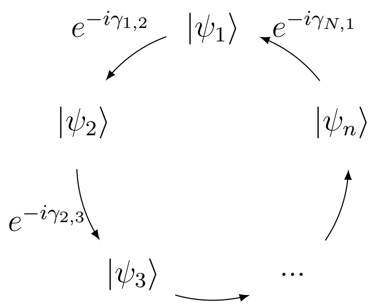
Again, the physical meaning of each state is unaffected, so we expect that any physical quantity calculated from these states must be invariant under the transformation. Thus one might expect that any *phase-like* quantity calculated from these states should have no physical relevance. For example, let us consider the relative phase between state j and k, $\gamma_{j,k}$, defined according to

$$e^{-i\gamma_{j,k}} = \frac{\langle\psi_j|\psi_k\rangle}{|\langle\psi_j|\psi_k\rangle|}. \tag{3.1.3}$$

Clearly this quantity is invariant if all states pick up the same phase, however it is not invariant under a local transformation, changing as

$$e^{-i\gamma_{j,k}} \to e^{-i(\gamma_{j,k}+\alpha_j-\alpha_k)}, \tag{3.1.4}$$

and so cannot be a physical observable. In order to construct a gauge-invariant object, let's pick a set of n (≥ 3) states $|\psi_j\rangle$ and arrange them in a loop.



We can now define a new quantity called the Berry phase:

$$e^{-i\gamma_B} = \frac{\langle\psi_1|\psi_2\rangle\langle\psi_2|\psi_3\rangle \ldots \langle\psi_N|\psi_1\rangle}{|\langle\psi_1|\psi_2\rangle\langle\psi_2|\psi_3\rangle \ldots \langle\psi_N|\psi_1\rangle|} \tag{3.1.5}$$

Another way of writing this would be

$$\gamma_B = -\arg \operatorname{Tr}\left[\hat{\psi}_1\hat{\psi}_2 \ldots \hat{\psi}_N\right], \tag{3.1.6}$$

where the trace is over the whole system, and $\hat{\psi}$ is shorthand for the density operator

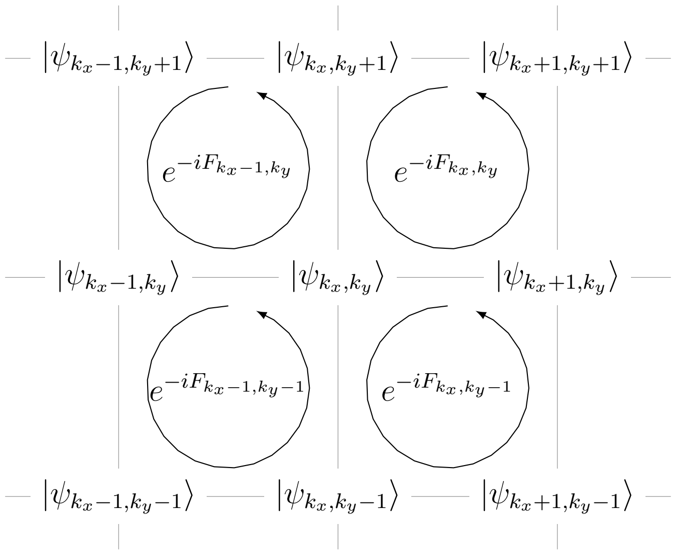
$$\hat{\psi}_i = |\psi_i\rangle\langle\psi_i|. \tag{3.1.7}$$

This quantity is clearly gauge-invariant, since any phase applied to a state $|\psi_m\rangle$ will show up twice, once from $\langle\psi_m|\psi_{m+1}\rangle$ and once from $\langle\psi_{m-1}|\psi_m\rangle$. The phase will have opposite sign in each of these expressions so they cancel out. Furthermore, we can see that the Berry phase around a loop is simply the sum of all the relative phases around that loop modulo 2π.

$$\gamma_B = \sum_{n=1}^{N} \gamma_{n,n+1} \mod 2\pi \tag{3.1.8}$$

Berry Flux

Now suppose that we have a large number of states arranged in a 2D grid. These states $|\psi_{\mathbf{k}}\rangle$ are labelled according with a pair of indices k_x and k_y. In general, this parameter $\mathbf{k}$ will refer to the crystal momentum, however the construction is equally valid for any other parameter that defines a 2D grid of quantum states.



We introduce a new quantity called the Berry flux. This is defined as the Berry phase around a plaquette on our grid,

$$F_{\mathbf{k}} = -\arg \mathrm{Tr}\left[\hat{\psi}_{k_x,k_y}\hat{\psi}_{k_x+1,k_y}\hat{\psi}_{k_x+1,k_y+1}\hat{\psi}_{k_x,k_y+1}\right]. \tag{3.1.9}$$

$F_{\mathbf{k}}$ is just the sum (mod 2π) of the relative phases along each edge of the plaquette starting at point $\mathbf{k}$. Thus, the Berry phase around some arbitrary loop containing multiple plaquettes can be found by summing the value of $F_{\mathbf{k}}$ for every plaquette contained in that region. Each relative phase flips sign depending on which way round it is evaluated. Any internal edge is shared between two plaquettes and so will appear twice in the sum of Berry fluxes, with opposite signs. This means all internal edges have their contribution to the total phase cancelled out and we are left with the sum of the relative phases only around the boundary of our region.

$$\exp\left[-i\sum_{\substack{\text{inside}\\\text{region}}} F\right] = \exp\left[-i\sum_{\text{boundary}}\gamma\right] \tag{3.1.10}$$

This resembles Stokes' law, however there is a subtlety worth mentioning. All we have required is that the complex exponential of each sum is the same. This does **not** mean that the sum of the Berry flux inside a region is equal to the sum of relative phases around the boundary, since they are allowed to differ by an arbitrary multiple of 2π.

Chern Number

Now let's assume that our grid of $\mathbf{k}$ points has periodic boundaries, with $1 \le k_x \le N_x$ and $1 \le k_y \le N_y$. That is, the lattice is a discretisation of the torus – a closed manifold. The product of all the Berry flux phases is now always equal to 1,

$$\prod_{k_x=1}^{N_x}\prod_{k_y=1}^{N_y} e^{-iF_{k_x,k_y}} = 1. \tag{3.1.11}$$

Here because every edge in our grid is now an internal edge and thus they cancel out. Again, as before, this does not mean that the sum of all Berry fluxes in our system vanishes. It means that the sum must always equal some multiple of 2π. This sum gives us the Chern number, $\mathcal{Q}$,

$$\mathcal{Q} = \frac{1}{2\pi} \sum_{k_x,k_y} F_{k_x,k_y}. \tag{3.1.12}$$

$\mathcal{Q}$ is necessarily restricted to integer values, since a non-integer value would contradict equation 3.1.11.

3.1.2 Topology of a General Hamiltonian

Now that we have a basic intuition for what the Chern number is, let's see how this applies to a real physical system. We start by defining some arbitrary Hamiltonian $H(\mathbf{R})$, which depends on some smooth parameter $\mathbf{R} \in \mathcal{R}$.

For every value of $\mathbf{R}$, the Hamiltonian will generally have some set of eigenstates $|\psi_n(\mathbf{R})\rangle$ with corresponding energies ε_n. As before, these eigenstates are only strictly defined up to a phase. Thus let consider a single state $|\psi(\mathbf{R})\rangle$ and define a local gauge transformation as applying some $\mathbf{R}$-dependent phase $\alpha(\mathbf{R})$ to all of the states over this band.

$$|\psi(\mathbf{R})\rangle \to e^{-i\alpha(\mathbf{R})} |\psi(\mathbf{R})\rangle . \tag{3.1.13}$$

In order for this to be well-behaved we will need to use a *smooth gauge*. This is a choice of gauge that ensures that the $\mathbf{R}$-derivative of a state, $\nabla_{\mathbf{R}} |\psi(\mathbf{R})\rangle$ is always well-defined – that is, $\alpha(\mathbf{R})$ does not jump around discontinuously. In general it is not always possible to pick a gauge that is globally smooth over all of $\mathcal{R}$, however it *is* always possible to pick a locally smooth gauge around the neighbourhood of some point in $\mathcal{R}$.

Let us define relative phase between a state at $\mathbf{R}$ and another state that is infinitesimally displaced from it, at $\mathbf{R} + d\mathbf{R}$,

$$e^{-i\Delta\gamma} = \frac{\langle\psi(\mathbf{R})|\psi(\mathbf{R}+d\mathbf{R})\rangle}{|\langle\psi(\mathbf{R})|\psi(\mathbf{R}+d\mathbf{R})\rangle|}. \tag{3.1.14}$$

In the limit of small $d\mathbf{R}$ we can express $|\psi(\mathbf{R}+d\mathbf{R})\rangle$ to first order in $d\mathbf{R}$ as

$$|\psi(\mathbf{R}+d\mathbf{R})\rangle = |\psi(\mathbf{R})\rangle + d\mathbf{R}\cdot\nabla_{\mathbf{R}} |\psi(\mathbf{R})\rangle . \tag{3.1.15}$$

Which means that

$$\langle\psi(\mathbf{R})|\psi(\mathbf{R}+d\mathbf{R})\rangle = 1 + \langle\psi(\mathbf{R})|\nabla_{\mathbf{R}}|\psi(\mathbf{R})\rangle \cdot d\mathbf{R}. \tag{3.1.16}$$

Now, noticing that $\langle\psi(\mathbf{R})|\nabla_{\mathbf{R}}|\psi(\mathbf{R})\rangle$ is pure imaginary, we can show that

$$\Delta\gamma = \mathbf{A}\cdot d\mathbf{R}, \tag{3.1.17}$$

$$\mathbf{A}(\mathbf{R}) = i\,\langle\psi(\mathbf{R})|\nabla_{\mathbf{R}}|\psi(\mathbf{R})\rangle . \tag{3.1.18}$$

The quantity multiplying $d\mathbf{R}$ is the Berry connection $\mathbf{A}(\mathbf{R})$. Like the relative phase in the previous section, it is not a gauge-invariant quantity. Under a gauge transformation it changes according to

$$\mathbf{A}(\mathbf{R}) \to \mathbf{A}(\mathbf{R}) - \nabla_{\mathbf{R}}\alpha(\mathbf{R}) \tag{3.1.19}$$

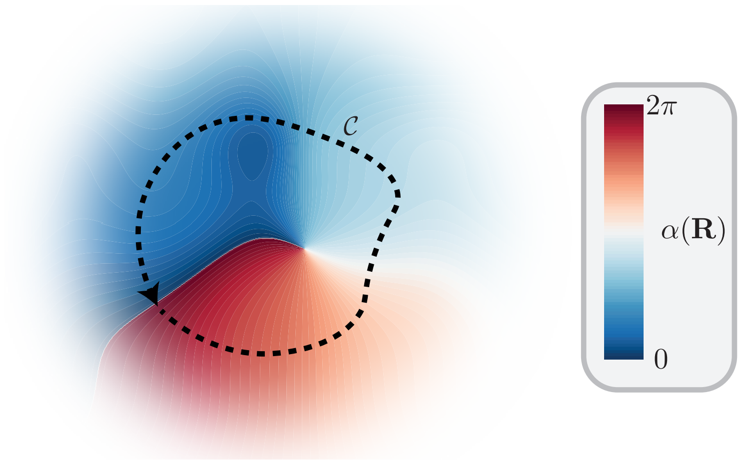


Figure 3.1: An example of a non-smooth gauge transformation designed to change the value of the Berry phase around the path $\mathcal{C}$ – marked with a dashed line – by an addition of 2π. Only a small region of $\mathbf{R}$-space is shown, with the local phase applied shown by the colormap.

Now let us suppose that we slowly transform our Hamiltonian around some closed path $\mathcal{C}$ in $\mathbf{R}$-space and keep track of the phase picked up by a given state.[1] In analogy with eqn. 3.1.8, the Berry connection allows us to calculate the total phase accumulated over this route, according to

$$\gamma(\mathcal{C}) = \oint_{\mathcal{C}} \mathbf{A} \cdot d\mathbf{R}. \tag{3.1.20}$$

The exponential of the phase $e^{-i\gamma(\mathcal{C})}$ around the loop *is* gauge invariant, one can see this intuitively by realising that it is identical to the Berry phase from the discrete case, but in the limit where the discrete steps become arbitrarily close together. However, the Berry phase $\gamma(\mathcal{C})$ itself is not strictly gauge invariant, since one could imagine introducing a gauge transformation that introduces a branch cut around the path $\mathcal{C}$. After traversing the whole path, the state at the end of the loop would have picked up a total phase 2π, meaning that under a gauge transformation we can change the berry phase according to

$$\gamma(\mathcal{C}) \to \gamma(\mathcal{C}) + 2\pi n \text{ for } n \in \mathbb{Z}. \tag{3.1.21}$$

Berry Curvature

Let us now define a quantity equivalent to the Berry flux from the previous discussion. Starting with eqn. 3.1.20, we use Stokes' theorem to transform it into a surface integral over a surface $\mathcal{S}$ bounded by the curve $\mathcal{C}$. We will consider the case for a two and three dimensional parameter

[1]Many sources (e.g. [2]) present this picture as an *adiabatic* transformation, where one has to do a little extra work to keep track of the energy-dependent phase that a state picks up $\sim e^{-i\varepsilon t}$ alongside the topological phase determined by $\mathbf{A}$. We won't bother, and instead just consider comparing the *instantaneous eigenstates* at different values of $\mathbf{R}$.

space $\mathcal{R}$, but of course this argument generalises to arbitrary dimensions.

$$\text{In 2D: } \gamma(\mathcal{C}) = \int_{\mathcal{S}} d^2R \left(\partial_x A_y(\mathbf{R}) - \partial_y A_x(\mathbf{R})\right), \tag{3.1.22}$$

$$\text{In 3D: } \gamma(\mathcal{C}) = \int_{\mathcal{S}} d\mathbf{S} \cdot \left(\nabla_{\mathbf{R}} \times \mathbf{A}(\mathbf{R})\right), \tag{3.1.23}$$

where we use the shorthand $\partial_\alpha = \frac{\partial}{\partial R_\alpha}$. Thus we may define the Berry curvatures,

$$\Omega(\mathbf{R}) = \partial_x A_y(\mathbf{k}) - \partial_y A_x(\mathbf{R}) \tag{3.1.24}$$

$$\boldsymbol{\Omega}(\mathbf{R}) = \nabla_{\mathbf{R}} \times \mathbf{A}(\mathbf{R}). \tag{3.1.25}$$

Now let us focus only on the 2D case. Writing this quantity out explicitly in terms of the states $|\psi(\mathbf{R})\rangle$, we arrive at

$$\Omega(\mathbf{R}) = i \left\langle \partial_x \psi(\mathbf{R}) | \partial_y \psi(\mathbf{R}) \right\rangle + h.c. \tag{3.1.26}$$

We can insert a resolution of the identity $\mathbb{1} = \sum_n |\psi_n\rangle\langle\psi_n|$ to obtain

$$\Omega(\mathbf{R}) = i \sum_{\psi_n \neq \psi} \left\langle \partial_x \psi | \psi_n \right\rangle \left\langle \psi_n | \partial_y \psi \right\rangle + h.c. \tag{3.1.27}$$

Using the identity,

$$\left\langle \psi_n | \partial_\alpha \psi \right\rangle = \frac{\left\langle \psi_n | \partial_\alpha H | \psi \right\rangle}{\epsilon - \epsilon_n}, \text{ for } \psi \neq \psi_n, \tag{3.1.28}$$

we can re-express the Berry curvature in the form

$$\Omega(\mathbf{R}) = i \sum_{\psi_n \neq \psi} \frac{\left\langle \psi | \partial_x H | \psi_n \right\rangle \left\langle \psi_n | \partial_y H | \psi \right\rangle}{(\epsilon - \epsilon_n)^2} + h.c. \tag{3.1.29}$$

This calculation has one important caveat. It is only possible to apply Stokes' theorem when $\mathbf{A}$ is smooth over the region $\mathcal{R}$. This means that we need to have already picked a smooth gauge in the region we are integrating over to have a guarantee that eqns. 3.1.22 and 3.1.22 hold. If we transform to a non-smooth gauge then it is possible that the Berry phase will change by a multiple of 2π, however the Berry curvature is truly gauge invariant. This can be seen from eqn. 3.1.29, where the Berry curvature is defined without the differentiation of states, so is invariant under even a non-smooth gauge transformation. Thus in general we only have the guarantee that

$$e^{-i\gamma(\mathcal{C})} = e^{\int_{\mathcal{S}} d^2R\, \Omega(\mathbf{R})}. \tag{3.1.30}$$

Chern Number

As in §3.1.1, we make the assertion that the $\mathbf{R}$-space lives on the torus, a closed manifold. The Chern number can then be defined as the integral of the Berry curvature across the whole manifold,

$$\mathcal{Q} = \frac{1}{2\pi} \int d^2 R \, \Omega. \tag{3.1.31}$$

As in the discrete case, the Chern number here is restricted to only take integer values. The argument follows the same lines as previously, where we can use our modified form of Stokes theorem (eqn. 3.1.30) to show that the Chern number is equivalent to the Berry phase around a loop enclosing no area, implying that

$$e^{\int d^R k \, \Omega(\mathbf{R})} = 1, \implies \mathcal{Q} \in \mathbb{Z} \tag{3.1.32}$$

3.1.3 An Example – The Two-Level System

Let us illustrate these concepts with the example of a two-state system. Our first step is to parametrise the possible Hamiltonians in such a system using a vector $\mathbf{d}$. We will then solve this Hamiltonian for the positive and negative energy eigenstates. This allows us to look at how the eigenstates change as we smoothly deform the Hamiltonian, exploring the parameter space of $\mathbf{d}$. By examining how the eigenstates of $\hat{H}$ depend on the parametrisation, we can calculate the Berry curvature of the positive and negative energy states. We then connect this with our understanding of the Bloch states in a periodic quantum system. We discuss how the bulk Hamiltonian of a periodic system corresponds to the embedding of a closed surface in $\mathbf{d}$-space and describe how to calculate the Chern number of the bands in such a system.

The Hilbert space of a two-level system is $\mathbb{C}^2$. This means that any Hamiltonian we construct must be a 2×2 Hermitian matrix. We will parametrise this in terms of a Pauli vector, $\mathbf{d}$,

$$H = \mathbf{d} \cdot \hat{\sigma} = d_i \sigma^i. \tag{3.1.33}$$

where the σ^i are the Pauli matrices, which form a complete basis for 2×2 traceless hermitian matrices. Of course, we could include a multiple of the identity to span the full space of Hermitian matrices, however this would have no effect on the topology so we shall omit it.

The set of traceless Hamiltonians is thus a manifold in itself, $\{H\} \cong \mathbb{R}^3$, since $\mathbf{d} \in \mathbb{R}^3$. The first thing we will do is to express the vector $\mathbf{d}$ in standard spherical polar coordinates, giving us a Hamiltonian that looks like

$$H = |\mathbf{d}| \begin{pmatrix} \cos(\theta) & e^{-i\phi} \sin\theta \\ e^{i\phi} \sin\theta & -\cos(\theta) \end{pmatrix} \tag{3.1.34}$$

This matrix has $H^2 = |\mathbf{d}|^2 \mathbb{1}$, so its eigenvalues are $\pm|\mathbf{d}|$. One can show that its eigenvectors are

$$|+_\mathbf{d}\rangle = \begin{pmatrix} e^{-i\phi/2} \cos(\theta/2) \\ e^{i\phi/2} \sin(\theta/2) \end{pmatrix}, \; |-_\mathbf{d}\rangle = \begin{pmatrix} -e^{-i\phi/2} \sin(\theta/2) \\ e^{i\phi/2} \cos(\theta/2) \end{pmatrix}. \tag{3.1.35}$$

We have thus arrived at a set of states, $|\pm_\mathbf{d}\rangle$ that depend on some parameter $\mathbf{d}$. This means that we can apply the techniques of the last section. From eqns. 3.1.18 and 3.1.23 we know that the Berry connection and Berry curvature of the $|\pm\rangle$ states are given by

$$\mathbf{A}^{\pm}(\mathbf{d}) = i \langle \pm_\mathbf{d} | \nabla_\mathbf{d} | \pm_\mathbf{d} \rangle \tag{3.1.36}$$

$$\mathbf{\Omega}^{\pm}(\mathbf{d}) = \nabla_\mathbf{d} \wedge \mathbf{A}^{\pm}(\mathbf{d}). \tag{3.1.37}$$

Using this equation it is possible to directly calculate the Berry curvature, arriving at the following expression:

$$\boldsymbol{\Omega}^{\pm} = \pm \frac{\hat{n}_{\mathbf{d}}}{2\mathbf{d}^2} \tag{3.1.38}$$

where $\hat{n}_{\mathbf{d}}$ is a unit vector pointing in the direction of $\mathbf{d}$.

The Berry curvature is equivalent to a monopole field radiating out from the origin in $\mathbf{d}$-space, in analogy with electrostatics. Thus, the Berry phase of the $|+_{\mathbf{d}}\rangle$ state around some path in $\mathbf{d}$-space is given by

$$\gamma^{+}(C) = \oint_{\mathcal{C}} \mathbf{A}^{+}(\mathbf{d}) \cdot d\mathbf{d}. \tag{3.1.39}$$

Re-expressing this in terms of the Berry curvature gives us

$$\gamma^{+}(C) = \int_{\mathcal{R}} \boldsymbol{\Omega}^{+}(\mathbf{d}) \cdot d\mathbf{S}, \tag{3.1.40}$$

where the integral is over a region $\mathcal{R}$ that has the curve $\mathcal{C}$ as its boundary. Since the Berry curvature is just a monopole field with charge $\frac{1}{2}$, this integral ends up evaluating half the solid angle subtended by the curve $\mathcal{C}$.

Chern Number

Now we can connect this with our understanding of Bloch states. For a two-dimensional translationally symmetric system, the Hamiltonian can be diagonalised according to

$$H = \sum_{\mathbf{k}} |\mathbf{k}\rangle \langle \mathbf{k}| \otimes H(\mathbf{k}). \tag{3.1.41}$$

If the unit cell contains two sites, then $H(\mathbf{k})$ is a 2×2 matrix that depends on $\mathbf{k}$. We can write it in terms of a $\mathbf{d}$ vector that depends on $\mathbf{k}$.

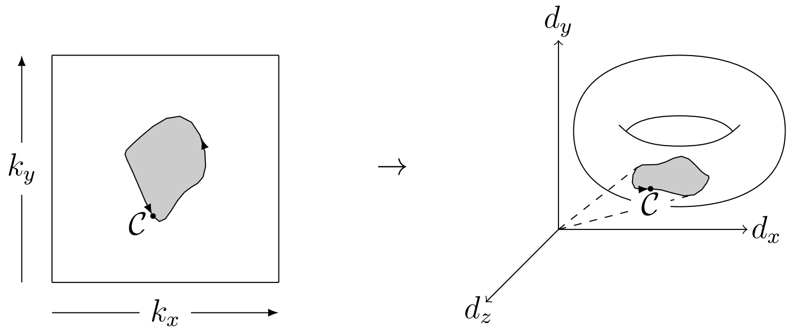
$$H(\mathbf{k}) = \mathbf{d}(\mathbf{k}) \cdot \sigma. \tag{3.1.42}$$

$\mathbf{k}$ lives on the surface of a torus, since k_x and k_y are periodic. Furthermore, $\mathbf{d}$ (and by extension H) live in the space $\mathbb{R}^3$. Thus, the mapping $\mathbf{d}$ is

$$\mathbf{d} : S^1 \times S^1 \to \mathbb{R}^3. \tag{3.1.43}$$

The map is embedding a two-dimensional surface (the torus) into three-dimensional flat space ($\mathbb{R}^3$). There are no limits on how you can do this embedding, the torus can take any shape provided the embedding respects the topology of the Brillouin zone.

To calculate the Berry phase around a loop in k-space, all we have to do is look at the corresponding loop in d-space. Then we integrate the Berry connection over some surface that has this loop as its boundary. The Berry connection actually looks like a radiating monopole so we can simplify further, knowing that we just need to find the solid angle subtended by the curve around the origin and divide it by two.



What about the Chern number? To calculate this all we need to do is integrate the Berry curvature across the whole surface of the torus in d-space. If the embedding of the torus does not contain the origin, then the value of this integral is necessarily zero.

Now suppose the torus is embedded such that it contains the origin, then we will have a surface that catches all of the Berry flux on its inside and the solid angle subtended is 4π! This means that the Chern number will be +1. Furthermore, you could imagine wrapping the torus around the origin twice, so that it effectively contains two origins, now the Chern number will be +2. You could turn it inside out and then wrap it around the origin too, that would give $\mathcal{Q} = -1$ and so on...

3.1.4 Final Notes

We close this section with an explanation of the consequences of Chern number in topological insulators.

Smoothly Deforming the Hamiltonian

Now suppose that we have a pair of quantum systems that live in the same Hilbert space, described by $\hat{H}_1$ and $\hat{H}_2$. Suppose both of these Hamiltonians represent insulating materials. This means that the bands in these materials are separated across all of the Brillouin zone by a band gap. We might want to ask the question: *Is it possible to smoothly deform $\hat{H}_1$ into $\hat{H}_2$ without ever crossing a point where the bands touch and the material becomes conductive?.*

The answer to this is that it is only possible if the bands of $\hat{H}_1$ and $\hat{H}_2$ have the same Chern numbers. This is because the Chern number is always restricted to have integer value, it cannot continuously change as we smoothly deform the Hamiltonian. In order to change the Chern number of a band we must somewhere cross a point where it is impossible to define a Chern number at all, allowing it to jump discontinuously when we leave that point. This is precisely what happens when two bands touch, the description of a single band becomes ill-defined and our calculation for Chern number stops working.

We can illustrate this with the example in §3.1.3. Smoothly deforming our Hamiltonian amounts to changing the shape of the torus embedded in **d**-space. If any point on the surface of the torus touches the origin, we have a pair of zero-energy states appearing in our system, this means the bands touch and the system is conductive. If the band has a Chern number of one, this means that the torus contains the origin. If we wanted to smoothly deform this to have zero Chern number, we would need to move the torus such that the origin was now outside. At some point in this deformation the torus surface will have to touch the origin, creating a conducting Hamiltonian at that point.

Thus we can classify all Hamiltonians according to the Chern number of their bands. It is always impossible to smoothly deform a Hamiltonian into one with a different Chern number while remaining insulating. Similarly, it is always possible to smoothly deform two Hamiltonians into one another without closing a gap, provided that their bands have identical Chern numbers.

This allows us to make a rough justification for why conducting edge states must appear wherever we have a material with nonzero Chern number. Suppose we have a large material where the Hamiltonian is allowed to vary slowly over space, defined by $\hat{H}(\mathbf{r})$. Now let's stay that in some region this Hamiltonian looks locally like it has non-zero Chern number, $\hat{H}_{\text{topological}}$, but in a different, distant region it looks locally like it has zero Chern number, $\hat{H}_{\text{trivial}}$. We expect that between these regions the Hamiltonian must be gradually transforming between $\hat{H}_{\text{topological}}$ and $\hat{H}_{\text{trivial}}$. As we have described, it is impossible for our Hamiltonian to make this transformation without creating a conducting surface somewhere between these two regions – edge states must appear! Loosely speaking, we can view the vacuum as a zero Chern number material, which gives us an intuitive picture for why we might expect materials with non-zero Chern number to always have edge states appearing on their surface, where the Hamiltonian has to transition to the vacuum.

Chern Number and Globally Smooth Gauges

If you thought carefully about the gauge that we chose in equation 3.1.35 you might notice that it has a problem, There is actually a discontinuity at $\phi = \pi$ where $e^{i\phi}$ suddenly jumps as you cross over to $\phi = -\pi$. This is no oversight, it turns out that it is actually completely impossible to define a globally smooth set of states $|\psi(\mathbf{d})\rangle$ for any band that has nonzero Chern number!

This is because if $|\psi(\mathbf{d})\rangle$ were continuous everywhere, then the Berry connection would also have to be continuous everywhere. This would mean that we could invoke the exact Stokes' theorem when calculating the Chern number. Since the Berry curvature is integrated is over the whole of $\mathbf{k}$-space it must have no boundary, $\mathcal{Q}$ must be equal to zero.

In this sense, the Chern number can be regarded as something like a measure of the amount of discontinuity that *must* be present regardless of the gauge you choose for the space.

3.2 The QWZ Model

Let us introduce the Qi-Wu-Zhang (QWZ) model as a paradigmatic example of a two-dimensional topological insulator with nearest-neighbour interactions [3, 4].

add some history on the model here

We work on a square lattice of unit cells indexed by a two dimensional lattice vector $\mathbf{m}$, where each cell contains two sites. The QWZ model is defined by three terms, a hopping between unit cells in the x-direction, a hopping in the y-direction, and a staggered on-site potential whose

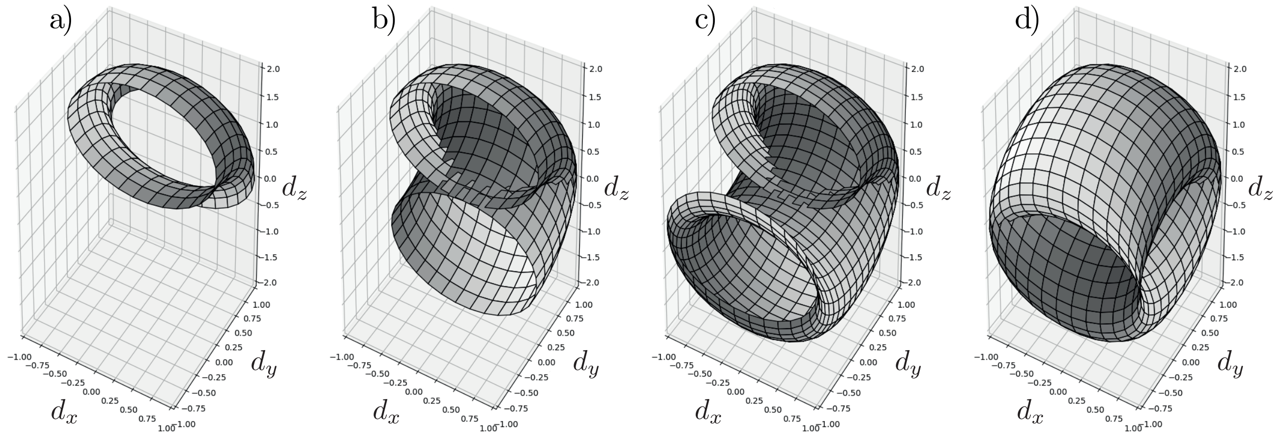


Figure 3.2: The embedding of the Brillouin zone in **d**-space for the QWZ model. This is shown in four progressive cutaways to ensure the internal structure can be seen. In all four images k_x spans the interval $[0, 2\pi)$, however in **(a)** k_y spans $\left[0, \frac{\pi}{2}\right)$, in **(b)** k_y spans $[0, \pi)$, in **(c)** k_y spans $\left[0, \frac{3\pi}{2}\right)$, and in **(d)** k_y spans $[0, 2\pi)$.

strength is determined by a parameter u.

$$\begin{aligned} \hat{H} &= \sum_{\mathbf{m}} \left|\mathbf{m} + \mathbf{1_x}\right\rangle \left\langle \mathbf{m}\right| \otimes \frac{\sigma_z + i\sigma_x}{2} + h.c. \\ &+ \sum_{\mathbf{m}} \left|\mathbf{m} + \mathbf{1_y}\right\rangle \left\langle \mathbf{m}\right| \otimes \frac{\sigma_z + i\sigma_y}{2} + h.c. \\ &+ \sum_{\mathbf{m}} \left|\mathbf{m}\right\rangle \left\langle \mathbf{m}\right| \otimes u\, \sigma_z, \end{aligned} \tag{3.2.1}$$

where 1_x and 1_y represent the unit displacement in their respective directions,

$$1_x = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \; 1_y = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{3.2.2}$$

Using Bloch's theorem, we can find the bulk Hamiltonian,

$$\hat{H}(\mathbf{k}) = \begin{pmatrix} \sin k_x \\ \sin k_y \\ u + \cos k_x + \cos k_y \end{pmatrix} \cdot \boldsymbol{\sigma}. \tag{3.2.3}$$

Here, k_x and k_y determine our position in the Brillouin zone. As explained in §3.1.3, this represents the embedding of a torus in **d**-space, shown in fig. 3.2.

The u-parameter has the effect of translating the torus by a fixed amount in the z-direction. From this we can see that the QWZ model only has non-zero Chern number for values of u between -2 and 2. Translating the torus by any more will move the origin completely outside. Thus the Chern number is

$$Q = \begin{cases} -1 & : \quad -2 < u < 0, \\ 1 & : \quad 0 < u < 2, \\ 0 & : \quad \textit{otherwise}, \end{cases} \tag{3.2.4}$$

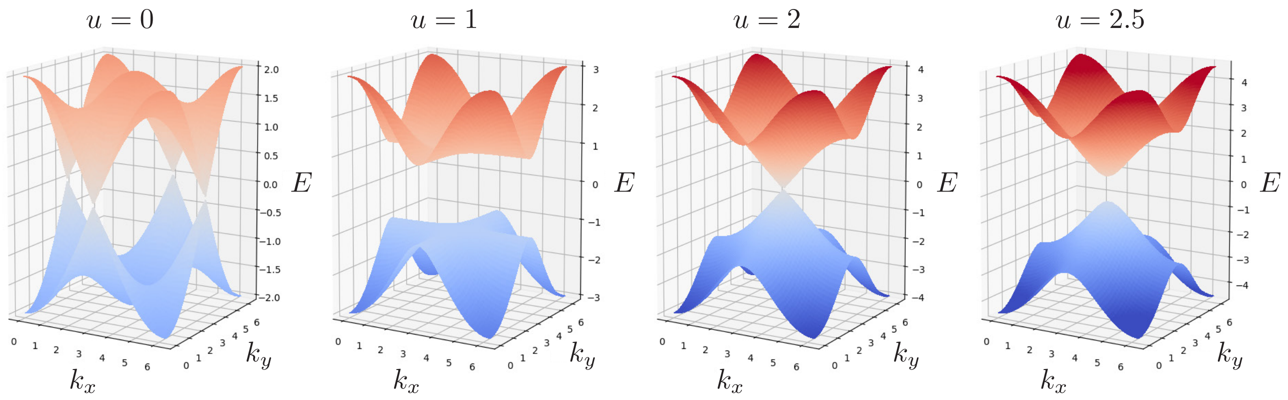


Figure 3.3: Four examples of the band structure of the QWZ model are shown, for varying u values between 0 and 2.5. At $u = 0$ and 2, the bands touch and Chern number is undefined. At $u = 1$, $\mathcal{C} = +1$, and at $u = 2.5$, $\mathcal{C} = 0$.

not including the points at $u =$ -2, 0 and 2, at which the Chern number is undefined because the band gap closes and the material becomes conducting. Some example band structures for varying values of u are shown in fig. 3.3.

3.2.1 Strip Geometry and Edge Modes

According to the bulk-boundary correspondence, whenever the Chern number $\mathcal{C}$ is nonzero, the QWZ model must have gapless edge modes appearing. To provide an example, here we construct the model on strip geometry and provide an analytical expression for the edge modes.

We shall require that the system is periodic and uniform in the y-direction, but is allowed to be non-uniform and have open boundaries in the x-direction. We allow u to vary in the x-direction, so add a subscript to indicate position-dependence, $u \to u_{m_x}$. We can apply Bloch's theorem in the y-direction only, and our eigenstates take the following form,

$$|\psi\rangle = |k_y\rangle \otimes |v_{k_y}\rangle\,, \tag{3.2.5}$$

where $|k_y\rangle$ is a plane wave in the y direction and $|v_{k_y}\rangle$ is a $2 \times L_x$ element vector. The v-states are solutions of the semi-bulk Hamiltonian

$$\begin{aligned}\hat{H}(k_y) = &\sum_{m_x} |m_x + 1\rangle \langle m_x| \otimes \frac{\sigma_z + i\sigma_x}{2} + h.c.\\ &+ \sum_{m_x} |m_x\rangle \langle m_x| \otimes \left[(u_{m_x} + \cos k_y)\, \sigma_z + \sin k_y\, \sigma_y\right].\end{aligned} \tag{3.2.6}$$

We have effectively reduced our system from a lattice of sites in x and y, to a set of chain models in x, that are each labelled by a value of k_y. The band structure for this system in both periodic boundaries and strip geometry is shown in fig. 3.4. Note that edge states are only present for certain values of k_y, which can be seen in figure 3.4, where the edge states peel away from the bulk in a region around $k_y = \pi$.

The strip Hamiltonian can be solved by proposing modes that decay exponentially away from

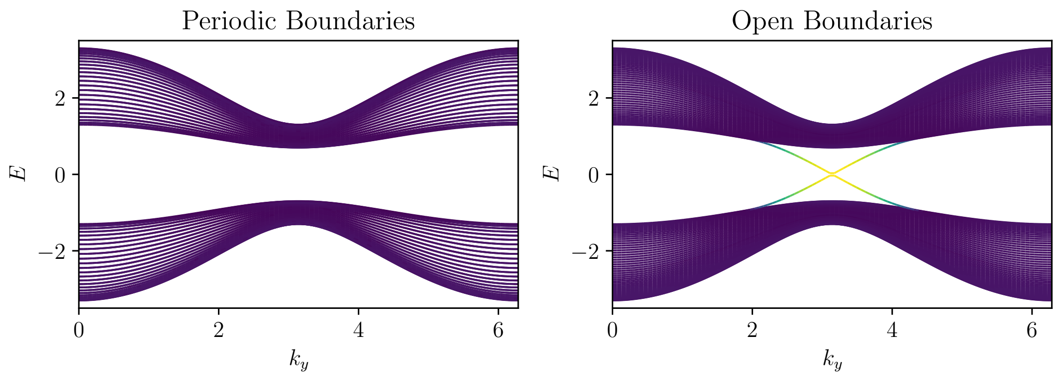


Figure 3.4: An example band structure for a QWZ model with $u = 1.3$ and $L_x = 30$. States are coloured according to their inverse participation ratio, indicating their relative degree of localisation. **(Left)** The example for periodic boundary conditions, showing two well-separated bands. **(Right)** The case for open boundary conditions. Two edge states are shown leaving the bulk and crossing the gap, with the descending state being a left edge state, and the ascending state being a right edge state.

the boundary. Thus, a left-edge state has the form

$$|\psi_{k_y}^{\text{left}}\rangle = \sqrt{1-\xi_l^2}\sum_{m=1}^{L_x}\xi_l^{m-1}|m\rangle \otimes \begin{pmatrix}1\\-i\end{pmatrix} \tag{3.2.7}$$

where

$$\xi_l = -u - \cos k_y. \tag{3.2.8}$$

This state has energy

$$E_l = -\sin k_y. \tag{3.2.9}$$

Similarly, the right edge state is determined by by

$$|\psi_{k_y}^{\text{right}}\rangle = \sqrt{1-\xi_r^2}\sum_{m=1}^{L_x}\xi_r^{L_x-m}|m\rangle \otimes \begin{pmatrix}1\\i\end{pmatrix} \tag{3.2.10}$$

$$\xi_r = -u - \cos k \tag{3.2.11}$$

$$E_r = \sin k. \tag{3.2.12}$$

These edge states are only valid solutions for k_y in the region

$$-1 < u + \cos k_y < 1. \tag{3.2.13}$$

These limits can be seen in fig. 3.4, where the edge states peel away from the bands to cross the energy gap. These expressions are only strictly valid in the limit of a large system. since we have made the assumption the edge states can be calculated by taking into consideration only one edge and ignoring the effect of the other. The edge states decay exponentially through the system, so we are relying on the assumption that the edge state has effectively decayed

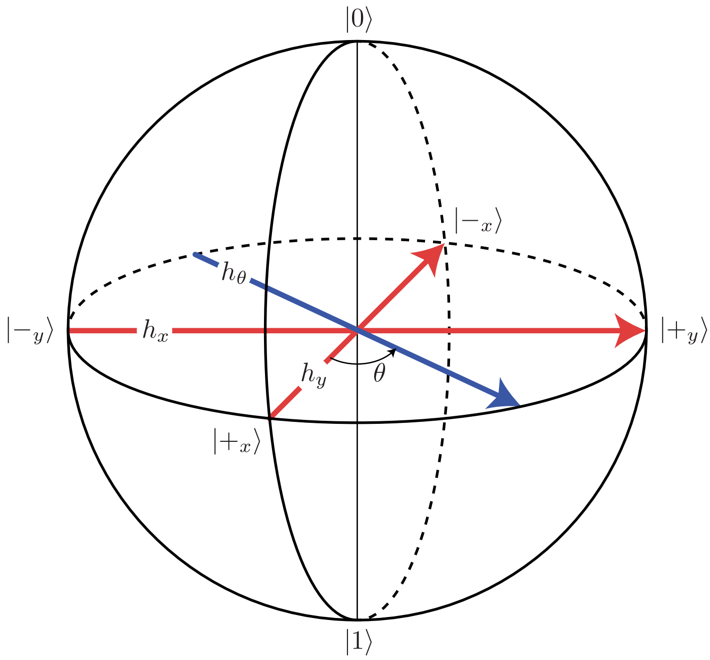


Figure 3.5: The Bloch sphere, showing the positions of the eigenstates of the three Pauli matrices. The action of the hopping matrices is shown in red, where the h_x and h_y operators act as creation operators on the y and x bases respectively. The action of our arbitrary angle hopping operator is shown in blue, acting as a creation operator between a pair of orthogonal states that lie on the x-y plane.

to nothing before reaching the other side. When this condition is relaxed the edge states are able to couple to one another. This has a two effects, the first being that at the crossing of edge states, our energy eigenvalues become positive and negative superposition states of both edge states. The other is that the left state coupling to the right state shifts their energy levels producing an avoided crossing such that no states cross $E = 0$.

3.2.2 The Amorphous QWZ Model

We now look at extending the QWZ model beyond a grid to arbitrary amorphous lattices. This is done by defining a smooth continuation of the QWZ model to allow for edges that can point in an arbitrary direction. Given an arbitrary lattice, we first require that every vertex hosts two states, and the on-site energy term containing $u\sigma_z$ is identical as in eqn 3.2.1. To construct an amorphous equivalent of the hopping terms, let us first look at the hopping in the QWZ model more carefully. We have two hopping matrices, one that hops a state in the x-direction, h_x, and one that hops in the y-direction, h_y.

$$h_x = \frac{\sigma_z + i\sigma_x}{2} = \frac{1}{2}\begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} \tag{3.2.14}$$

$$h_y = \frac{\sigma_z + i\sigma_y}{2} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}. \tag{3.2.15}$$

These operators can be viewed as creation operators on different bases, h_x acts on the basis of σ_y, and h_y acts on the basis of σ_x. Viewed in the Bloch sphere, these operators hop the state across the sphere in the x or y direction.

In our amorphous lattice, hopping operators can act across an edge connecting two sites that points in an arbitrary direction. Thus, rather than trying to assign edges with h_x or h_y operators, let us define an arbitrary-angle hopping matrix h_θ, that hops in a direction θ over the x-y plane of the Bloch sphere,

$$h_\theta = \left|+_\theta\right\rangle \left\langle -_\theta\right| \tag{3.2.16}$$

$$= \frac{1}{2}\begin{pmatrix} 1 & -e^{-i\theta} \\ e^{i\theta} & -1 \end{pmatrix}. \tag{3.2.17}$$

Thus, the Hamiltonian we use for the amorphous system is given by

$$\begin{aligned} \hat{H} = &\sum_{i,j} \left|\mathbf{r_i}\right\rangle \left\langle \mathbf{r_j}\right| \otimes h_{\theta_{i,j}} + h.c. \\ &+ \sum_{i} \left|\mathbf{r_i}\right\rangle \left\langle \mathbf{r_i}\right| \otimes u_i \sigma^z, \end{aligned} \tag{3.2.18}$$

where $\theta_{i,j}$ is the angle between the edge connecting $\mathbf{r_i}$ and $\mathbf{r_j}$ and the x axis.

As before, the parameter u controls whether the system is in a topological or trivial state. The exact dependence of the system on u depends strongly on the lattice geometry, however there is generally a region around which the system has nonzero topological character. In the absence of a momentum-space description, the system's topological character can be seen indirectly by comparing the spectrum in open and periodic boundary conditions. Generally one expects the presence of a gap in periodic boundaries that becomes populated with edge modes in open boundaries. Two examples are shown in fig. 3.6.

OPTIONAL – you could add a section here on the low-k theory of the amorphous QWZ model if you wanted - would allow you to make some predictions for the Chern number as you can find the berry phase change around the low k band touching. Maybe its a bit overkill - the thesis looks like its gonna be kinda long anyway.

References

[1] D. J. Thouless, M. Kohmoto, M. P. Nightingale, and M. den Nijs, "Quantized Hall Conductance in a Two-Dimensional Periodic Potential", Physical Review Letters **49**, 405–408 (1982).

[2] D. Xiao, M.-C. Chang, and Q. Niu, "Berry phase effects on electronic properties", Rev. Mod. Phys. **82**, 1959–2007 (2010).

[3] X. L. Qi, Y. S. Wu, and S. C. Zhang, "Topological quantization of the spin Hall effect in two-dimensional paramagnetic semiconductors", Physical Review B **74**, 085308 (2006).

[4] J. K. Asbóth, L. Oroszlány, and A. Pályi, *A Short Course on Topological Insulators: Band Structure and Edge States in One and Two Dimensions*, Lecture Notes in Physics (Springer International Publishing, 2016).

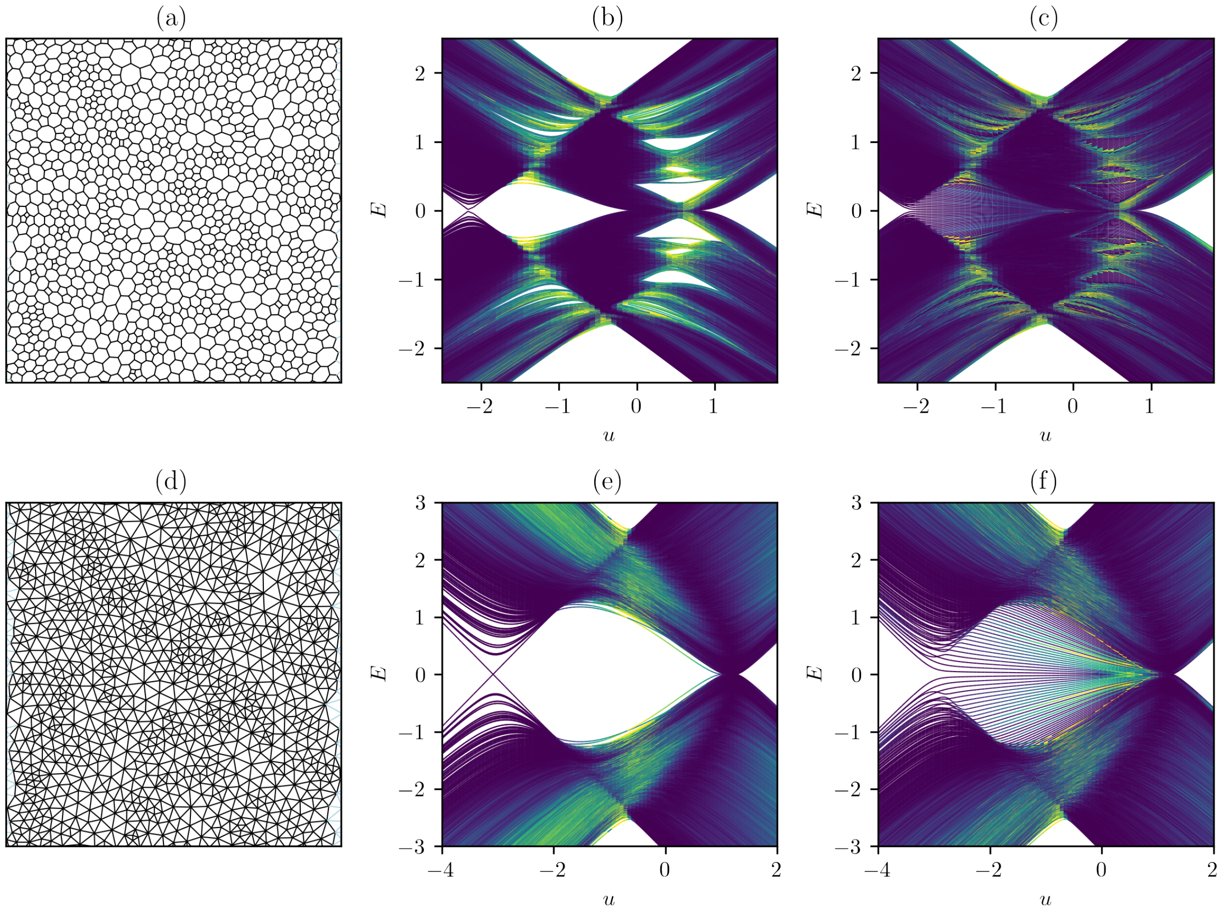


Figure 3.6: Plots showing the spectra of the amorphous QWZ model for two different example lattices. States are coloured according to their inverse participation ratio. **(a)** A trivalent lattice generated from a Voronoi decomposition of a set of random points, followed by five iterations of LLoyd's algorithm to regularise bond lengths. **(b)** The spectrum of the QWZ model on the voronised lattice as a function of u. Several gaps are present at different values of u and E. **(c)** The spectrum of the same model as (b), but in open boundary conditions. Here we can see that all gaps for $u \in [-1.5, 1]$ are now filled with edge modes, indicating that the system is in a topological phase. **(d)** An example of a lattice forming a Delaunay triangulation, constructed as the dual of the lattice in (a). **(e)** The spectrum of the QWZ model on the Delaunay lattice as a function of u showing one single large gapped region. **(f)** The spectrum of the same model as (e), but in open boundary conditions. Here we can see that the gaps is now filled with edge modes, again indicating that here the system is in a topological phase.

Chapter 4

Local Markers

4.1 Kitaev's Local Marker

4.2 The Bott Index

4.3 The Chern Marker

4.4 Finite Size Effects

Chapter 5

The Crosshair Marker

Chapter 6

Kitaev's Honeycomb Model

Chapter 7

The Amorphous Kitaev Model

Chapter 8

The Star Lattice Kitaev-Heisenberg Model

Chapter 9

Conclusion and Outlook